Ademola Adeigbe

Software & AI Engineer | Full-Stack Applications, RAG and ML Systems

London, United Kingdom | Relocating to Berlin: January 2027
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PROFILE

Software engineer with an MSc in Artificial Intelligence & Data Science (Merit), building full-stack applications and AI systems and measuring how well they work. Recent work includes StepFree London, a live Spring Boot and React journey planner with 288 automated tests, and a hybrid RAG pipeline whose top-result retrieval accuracy rose from 53% to 74%. Brings two years of commercial web development with React and Next.js, plus backend experience in Java and Python.

AI AND ML PROJECTS

UK Visa RAG Assistant | Python, Claude API, BM25, Sentence Transformers, FastAPI, Streamlit

- Built a RAG assistant over official gov.uk guidance, combining BM25 and dense embeddings with reciprocal rank fusion.
- Added cross-encoder reranking, raising hit@1 from 0.53 to 0.74 and hit@3 from 0.75 to 0.95 on a 100-question evaluation, for about 95 ms of extra CPU latency.
- Compared two reranker models across four candidate depths, and selected the smaller model because it was both more accurate and faster.
- Enforced source citations and refused to answer when retrieval confidence was low.

University IT Support RAG (MSc dissertation) | Python, scikit-learn, BM25, Ollama, Mistral 7B, FastAPI

- Built a cited IT-support assistant using an intent classifier, BM25 retrieval and a locally hosted Mistral 7B model.
- Declined 8 of 10 held-out out-of-scope questions where BM25 alone declined none, trading hit@3 from 0.97 to 0.90.
- Ran a component ablation across a 72-query set, which showed the rule layer contributed more to out-of-scope rejection than the classifier.
- Assessed generated answers with RAGAS and manual review of relevance, faithfulness and citation accuracy.

Fraud Detection MLOps | Python, XGBoost, FastAPI, MLflow, Docker, Prometheus, Grafana

- Trained an XGBoost model on 284,807 card transactions with 0.17% fraud, reaching PR-AUC 0.882, precision 0.87 and recall 0.84.
- Served predictions through FastAPI at 0.67 ms median and 1.1 ms p99 per request in a repeatable local benchmark.
- Implemented PSI drift detection, Prometheus and Grafana monitoring, and shadow-mode champion/challenger promotion, exercised with simulated traffic.
- Automated tests and Docker image publishing to GHCR with GitHub Actions, and wrote Kubernetes manifests and Terraform for AWS EKS.

SOFTWARE ENGINEERING PROJECTS

StepFree London | Java 21, Spring Boot, React, TypeScript, PostgreSQL, Docker

- Built accessible journey planning and disruption alerts on the TfL Unified API, showing missing accessibility data as unavailable rather than guessing.
- Covered the system with 288 automated backend, frontend and browser tests, including WCAG 2.1 AA accessibility checks.
- Deployed it as a single Docker image on Render with Neon PostgreSQL, live at stepfree-london.onrender.com.

Enterprise Service Management Platform | Java 21, Spring Boot, PostgreSQL, React, Docker

- Built an IT service desk with role-based access, workflows defined as data, and a complete audit trail.
- Implemented JWT access tokens with rotating refresh tokens and reuse detection that revokes the whole session family.

EXPERIENCE

Software Developer | Doyse Educational Consults | Remote | 10/2022 – 01/2025

- Developed and supported a live educational consultancy website with React and Next.js (doyseedu.com).
- Turned requirements from staff, content and design stakeholders into usable web features.
- Integrated APIs and reusable frontend components, and investigated issues on live pages.

Junior Web Developer Intern | Lagos State University Teaching Hospital | Lagos, Nigeria | 05/2022 – 10/2022

- Helped build and maintain web pages and the staff website with HTML, CSS, JavaScript and React.
- Investigated frontend and API issues by reproducing problems and tracing data flow.

Retail Supervisor | Jumia Nigeria | Nigeria | 09/2023 – 07/2025

- Coordinated day-to-day retail operations and supported customer-facing teams.

EDUCATION

MSc Artificial Intelligence & Data Science, Merit | University of Hull | 2025 – 2026

- Dissertation: A RAG-based Small Language Model System for University IT Support Query Resolution.

BSc Computer Science, Second Class Upper | Fountain University, Osogbo, Nigeria | 2019 – 2023

- GPA: 3.8/5.0.

SKILLS

- **AI and retrieval:** RAG, hybrid search, cross-encoder reranking, retrieval and answer evaluation, RAGAS, Claude API, local inference with Ollama
- **Machine learning:** XGBoost, scikit-learn, Pandas, NumPy, MLflow, drift detection (PSI)
- **Engineering:** Python, FastAPI, Java, Spring Boot, TypeScript, React, Next.js, SQL, PostgreSQL
- **Delivery:** Docker, GitHub Actions, Prometheus, Grafana, Kubernetes manifests, Terraform, Git

LANGUAGES

English: Fluent | German: A1 (actively learning)